

Splunk SIEM Lab – Step by Step Tutorial

This is a hands-on walkthrough of how I set up my **Splunk SIEM lab** from scratch on a Linux VM. The goal is to collect logs, build some dashboards, and play around with detections like a mini SOC environment.

◆ Step 1: Create Splunk User and Download

```
# Create splunk user and home dir
sudo useradd splunk -m -d /opt/splunk -s /bin/bash

# Switch to /tmp for download
cd /tmp

# Download Splunk Enterprise
wget -O splunk.tgz 'https://download.splunk.com/products/splunk/releases/9.2.0/linux/splunk-9.2.0-Linux-x86_64.tgz'
```

◆ Step 2: Install Splunk

```
# Extract into /opt
sudo tar -xvzf splunk.tgz -C /opt/

# Go into Splunk bin
cd /opt/splunk/bin

# Start Splunk and accept license
sudo ./splunk start --accept-license
```

Set up your admin username and password when prompted. After that, open a browser and go to:

```
http://<your_vm_ip>:8000
```

◆ Step 3: Create Indexes

In the Splunk web UI:

- Go to **Settings** → **Indexes** → **New Index**
- Add:
 - `security`
 - `os_logs`
 - `network`

◆ Step 4: Add Linux Logs

Back on the VM, tell Splunk to monitor system logs.

Edit `inputs.conf`:

```
sudo nano /opt/splunk/etc/system/local/inputs.conf
```

Add:

```
[monitor:///var/log/auth.log]
index = security
sourcetype = linux_secure
```

```
[monitor:///var/log/syslog]
index = os_logs
sourcetype = syslog
```

Restart Splunk:

```
sudo /opt/splunk/bin/splunk restart
```

Now Splunk will start pulling in auth and syslog data.

◆ Step 5: Add Windows Logs (Optional)

If you have a Windows machine:

1. Download **Splunk Universal Forwarder**.
2. Configure it to send `WinEventLog:Security` to your Splunk server.

Example snippet for `inputs.conf` on Windows:

```
[WinEventLog://Security]
index = os_logs
```

◆ Step 6: Install Security Apps

From the Splunk UI:

- Go to **Apps → Find More Apps**
 - Install:
 - Splunk Security Essentials
 - Splunk CIM (Common Information Model)
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◆ Step 7: Run Some Searches

Failed SSH Logins:

```
index=security sourcetype=linux_secure "Failed password"
| stats count by src, user
| sort -count
```

Windows Failed Logons:

```
index=os_logs sourcetype=WinEventLog:Security EventCode=4625
| stats count by user, src_ip
```

◆ Step 8: Create Alerts

Brute Force SSH Alert:

```
index=security sourcetype=linux_secure "Failed password"  
| stats count by src  
| where count > 10
```

Set an alert to email you if >10 failed logins come from the same IP.

Admin Login Burst:

```
index=os_logs user=Administrator EventCode=4624  
| bin _time span=5m  
| stats count by _time, src_ip  
| where count > 3
```